##Storage Tools and Techniques

Objectives:

At the end of this episode, I will be able to:

1. Identify tools and techniques to troubleshoot operating system issues.

Exam Objective: *4.3 - Given a scenario, troubleshoot storage problems.*

Description: In this episode, the viewer will identify the tools and techniques used to troubleshoot, diagnose and resolve storage issues such as partitioning tools, Window's Disk Management, system logs, *net use*, *mount* and visual inspections.

- Errors
  - Boot errors
    - Corrupted boot sector
    - OS not found
  - Sector block errors
    - Data corruption
  - Read/write errors
    - Data corruption
  - Partition errors
    - Data corruption
- Failure
  - Failed drives
    - Visual inspection
    - Auditory inspection
  - Cache battery failure
    - Data loss
    - Data corruption
    - Visual inspection
  - Restore failure
    - Corrupted backup
  - Cache failure
    - Component failure
    - Cache disable
  - Multiple drive failure
  - OS not found
- Performance
  - Slow I/O performance
  - Page/swap/scratch file or partition
    - Disk space
  - Slow file access
    - Disk space utilization
  - Unsuccessful backup
    - Insufficient disk space
    - Permission issues

- Device access issues
  - Unable to mount the device
    - Permissions
    - Network connections
  - Drive not available
    - Drive failure
    - Cable issues
    - Loose connections
    - Misconfigured RAID/failed RAID
    - Controller/HBA failure
  - Cannot access logical drive
    - Misconfiguration
- Tools and techniques
  - Disk management
  - RAID and array management
  - System logs
  - Disk mounting commands
    - net use
    - mount
  - Monitoring tools
  - Visual inspections
  - Auditory inspections

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Not applicable if blank